

PSEUDO-HERON'S *CHEIROBALLISTRA*

ONE MORE ENGLISH TRANSLATION.

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Abstract:

As a by-product of my work on the *Cheiroballistra*. I present here a new commented English translation of the treatise. I have tried to keep as close as possible to the original Greek text, displacing proposed emendations to the footnotes, for the sake of clarity.

Carefully re-drawn colour copies of the oldest extant copy of the treatise has been inserted, side to side with the related sections of text.

THE TREATISE

The *Cheiroballistra* is a very short treatise, written in Attic Greek. It is clearly divided into five sections, each one of them followed by a diagram.

The four main manuscript sources for establishing text and diagrams of the *Cheiroballistra* are the following ones¹:

M: Codex Parisinus inter supplementa Graeca 607 (foll. 56^r-58^v). Bibliothèque Nationale, Paris.

F: Fragmenta Vindobonensia 120, olim 113 (foll. 12^r-14^v). Österreichische Nationalbibliothek, Vienna.

P: Codex Parisinus Gr. 2442 (foll. 68^v-70^v). Bibliothèque Nationale, Paris.

V: Codex Vaticanus Gr. 1164 (foll. 106^v-108^v). Biblioteca Vaticana, Rome.

The treatise has reached us seemingly incomplete. Only the components of one small arrow-shooting catapult (or should we say *ballista*²?) are described. The text lacks a foreword, which is present even in the driest treatises³, and, maybe, more instructions about how to assemble the machine out from the already described components.

From the kind of machine described, the *Cheiroballistra* could have been written at any time between the end of the first century A.D. to, at least the sixth century A.D. The oldest surviving copy, **M**, perhaps dating from the tenth century

¹ See VINCENT 1866, 22-26; WESCHER 1876, IX-XL and MARSDEN 1971, 9-15, for short descriptions and relationship among these manuscripts.

² About that, see IRIARTE 2003.

³ Like that of Biton, for example.

A.D., exhibits a distribution of text and diagrams in the last two sections which strongly suggests that **M**'s archetype was copied from a papyrus roll. That would draw the writing date closer to the earlier than to the later period.

The Heron named in the title of most manuscripts is currently identified with the famous Heron of Alexandria, who lived during the second half of first century A.D. and the first half of the following one. It is highly improbable that Heron of Alexandria wrote the *Cheiroballistra*, due to the great differences in the vocabulary used to name the same objects in our text and in his well-known treatise on artillery, the *Belopoeika*. Moreover, the very name given to the machine - 'cheiroballistra' - is a late word, not contemporary to that Heron. As the only occurrence of both words - 'Heron' and 'cheiroballistra' - in the treatise is in the title, it could well be that this title were added to a fragmentary title-less text by a Byzantine copyist⁴.

There are several editions of the Greek text, following in the main that made by Carl Wescher⁵. The first one is that of Prou, which contained also Latin and French translations⁶. Schneider also included a German translation with his edition and mischievous critical notes⁷. The latest, and more readily available, editions of the *Cheiroballistra* are those of Marsden and Wilkins⁸. Both include excellent English translations, but must be taken with some caution, because the edited Greek text is that 'corrected' by the authors, instead of just the original one, with the proposed changes appearing as footnotes.

Why then another English translation? As I have said before, Marsden and Wilkins felt free to change the text whenever it did not fit their preconceived ideas on the machine. For that reason, I have felt compelled to bring forward this new one, following closely what the original Greek text says. I am not a philologist, therefore I have relied mainly on all the preceding translations -English or not-, resorting occasionally to the original manuscripts for solving doubts.

⁴ SCHNEIDER 1906, 167-168.

⁵ WESCHER 1867, 123-134. It was really not the first edition of the *Cheiroballistra* (PROU 1877, 2, 4-5, 11-14, 18-20), but it is now universally accepted as the standard one.

⁶ PROU 1877, 116-149. Some redrawn diagrams, of assorted sources, are included. They come mainly from M, but there are some from P and the bizarre P³, P⁵.

⁷ SCHNEIDER 1906, 146-163. B/W photographs of all the diagrams in M and P are included. There is another German translation of the *Cheiroballistra*, rather technical than philological, and due to Baatz (GUDEA & BAATZ 1974, 69-72). It includes redrawn diagrams from the Wescher edition, basically those of M.

⁸ MARSDEN 1971, 212-217. No diagrams included. WILKINS 1995, 10-32. It includes redrawn diagrams from M, P and V, but only the M series is complete.

Pertinent explanations and proposed emendations to the text appear as endnotes. Words between <> are assured additions to the text, smaller words between brackets are additions to make the text more understandable.

The measurements are given in the text in feet and digits. Whether Greek or Roman it is not fully settled, but the difference is not truly relevant. Both, Greek foot (*poys*) and Roman foot (*pes*) are subdivided into sixteen digits. A Greek digit (*daktylos*) amounts to 19.3 mm or 0.76 inches, while a Roman one (*digitus*) amounts to 18.5 mm or 0.73 inches.

The diagrams accompanying the translated sections are copies of those belonging to **M**, including the original coloring. Short commentaries complement each of the diagrams.

HERON'S CONSTRUCTION AND DIMENSIONS OF THE CHEIROBALLISTRA

Let two dovetailed boards be made, AB and $\Gamma\Delta$, with quadrilateral dovetails, of which let AB be the female⁹, $\Gamma\Delta$ the male¹⁰. Let AB have a length of 3 feet and 4 digits, a breadth of $3\frac{1}{2}$ digits, a thickness of $4\frac{1}{2}$ digits. Let $\Gamma\Delta$ have a length of 3 feet, a width of about $2\frac{1}{2}$ <digits>, a thickness of $1\frac{1}{4}$ digits.

Let the channel in the board AB have a depth of 1 digit. Let AZ , $2\frac{1}{2}$ feet <and> 6 digits (long), of AB have been channeled out. Thus remains $Z\Lambda$, 6 digits (long).

Again, let $A\Theta$ ¹¹, 1 foot and half and 4 digits (long), of the board AB be marked off, and AK , 1 foot and 1 digit (long). Thus remains $K\Theta$, 7 digits (long).

Again, let $1\frac{1}{2}$ digits be marked off from the thickness of $4\frac{1}{2}$ digits of the board AB , and let it be cut along AK and $A\Theta$, so that the portion $K\Theta$ remains at its original $4\frac{1}{2}$ digits; this is $X\Psi Y\Phi$.

Let a crescent-shaped piece HB also be made and, bored in the middle with a quadrangular hole, let it be (attached) firmly to the end AB of the board AB , as illustrated below.

Of the channel¹² $\Gamma\Delta$, let $E\Delta$ be a male dovetail and let it be suited to the female dovetail, the portion AZ of the channel¹³ AB ; that is the portion ΔE of the board $\Gamma\Delta$.

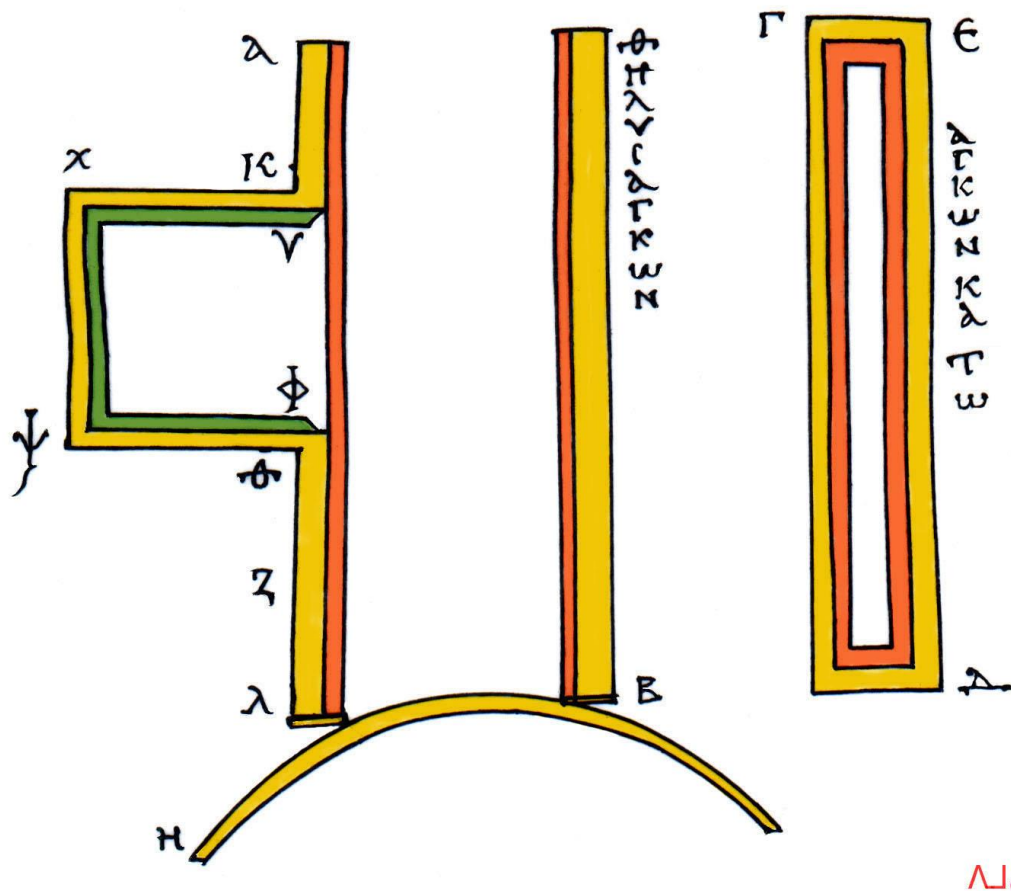
⁹ The case.

¹⁰ The slider.

¹¹ This should read $A\Theta$, but a copyist mistook the 'A' for an 'A', which is an easy error in the Greek uncial alphabet, and led to the mathematical nonsense: $28 \text{ d } (A\Theta) - 17 \text{ d } (AK) = 7 \text{ d } (K\Theta)$, as appears now in the text. Thus, $28 \text{ d } (A\Theta) + 17 \text{ d } (AK) + 7 \text{ d } (K\Theta) = 52 \text{ d } (A\Lambda = AB)$.

¹² The word 'channel' appears in all manuscripts, but 'board' would make more sense.

¹³ See the preceding footnote.



The first diagram depicts case and slider. Both are represented heavily stunted, and the case is difficult to recognize, at first sight, because significant traces of the ancient representation system still survive in its drawing. Basically, the board is seen from above, but only its side areas, not affected by the dovetailed channel, are figured as two narrow red stripes. Even though the point at which the channel must stop is marked, the grooved band reaches the case's lower end. It seems as if the two yellow stripes would represent the splayed side-elevations of the case, with the projecting block $X\Psi Y\Phi$ stemming from the underside of the left side-elevation (it does not appear in the right one, it is true, and the inner green stripe seemingly renders it as absurdly hollow!) The lunate fitting would be figured, horizontally laid, on the plan and, possibly, on the side-elevations too, as the two small yellow rectangles which are attached to the red and yellow stripes' lower ends.

The slider is figured, apparently, also seen from above as three concentric rectangles. Maybe they mean again the splayed side-elevations or maybe they mean just the slider's features as seen on a plan (i.e. the colourless inner rectangle would be the channel for the missile -unmentioned in the text-, the red middle one, the contour of the slider's upper surface and, the outer yellow one, the protruding contour of the male dovetail below.

II

Now we shall describe the trigger-mechanism.

Let a handle¹⁴ $AB\Gamma\Delta$ be made of iron, with the shape illustrated below; let there be a quadrangular tenon $Z\Theta$, with the part EZ as a double fork¹⁵; a bolt¹⁶ KAM ; a little snake¹⁷ $N\Xi$; an element in the form of a ‘pi’¹⁸ $O\Pi\P\Sigma$.

Let the handle $AB\Gamma\Delta$ be bored through Δ ; let the board $\Gamma\Delta$ in the first section be bored at MN , Ξ ¹⁹, at MN with a round hole right through, at Ξ with a quadrangular one. Let the handle be inserted in such a way that a pin, pushed through MN and through the hole Δ of the handle, establishes a bond.

Having bored holes through the double fork $E\Theta$ at TY and the bolt KAM at Φ , and having inserted a pin through both TY , Φ , we establish a bond, so that the bolt can move freely around it. Let the bolt have longitudinally an incision AM , 1 digit (long).

Having, thus, marked off ΔO , of 5 digits, on the board $\Gamma\Delta$ and bored a hole at O , we lower the double fork $E\Theta$ into it and establish a bond, so that it stays firm.

Next, boring through the little snake $N\Xi$ at N and the board $\Gamma\Delta$, the one in the first section, at Π , 4 digits from M ²⁰, we lower through the hole in the little snake and through Π a pin and establish a bond, so that the little snake $N\Xi$ can move easily around it.

Again, having marked off ΞP ²¹ from the handle $\langle AB \rangle \Gamma\Delta$, we bore a hole at P ; and again, having measured from there $4 \frac{1}{2}$ digits, (thus) $P\Sigma$, we bore a hole at Σ and then we lower the <element in the form of a ‘pi’> into the board $\Gamma\Delta$ ²², which appears in the first section.

Here it is.

¹⁴ *Cheiolabe*.

¹⁵ *Dichelon*.

¹⁶ *Schasteria*, i.e. the claw.

¹⁷ *Drakontion*, i.e. the trigger.

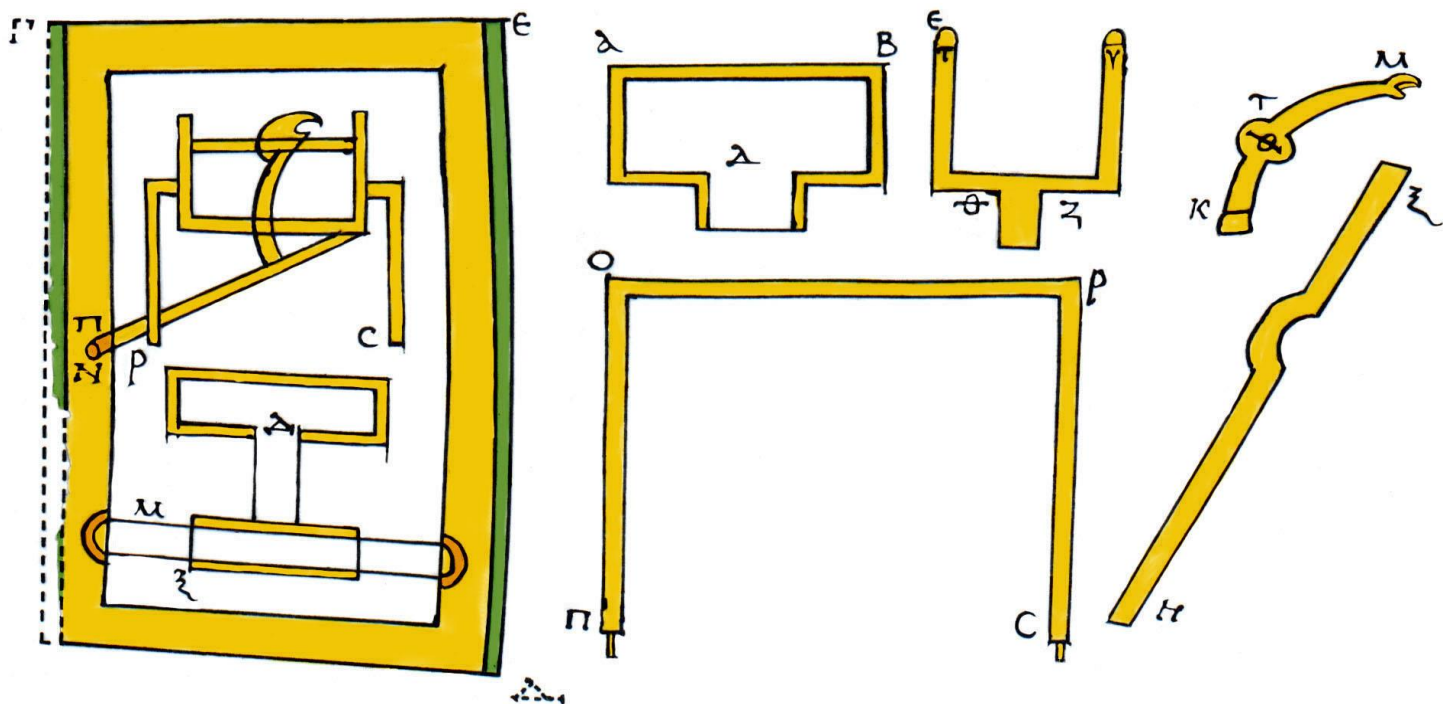
¹⁸ *Pittarion*. In the shape of the Greek letter ‘Π’.

¹⁹ Notice that the text does not give the exact relative positions of M , N and Ξ on the slider, we can only infer them approximately from the diagram.

²⁰ M of the pair MN , I suppose. Anyway, we do not know how far from Γ or Δ on the slider it is placed!.

²¹ Even if we suppose that Ξ is positioned at the Δ end of the slider, how long ΞP must be?

²² If we look at the diagram, $P\Sigma$ must be the width of the pittarion, but $4 \frac{1}{2} d$ is far wider than the slider itself is, about $2 \frac{1}{2} d$, so the paragraph must be corrupt.



The second diagram pictures the trigger mechanism, as assembled on the slider at the left side, and its individual components, at the right one.

The slider is once more represented viewed from above. It appears in the same configuration (more stunted, if possible) that we have previously seen, but here the positioning of the trigger mechanism shows unmistakably that the flat upper section is narrower than the dovetail below it. The 'bolt' (claw) is pivoting on the 'fork' and with its tail immobilised by the 'little snake' (trigger), placed below it. The 'pitarion' looks like placed behind the trigger, but the central section of its horizontal bar has been eliminated so as to not to obscure the fork-claw assembly. Finally, the 'handle' appears revolving around its axe, the bolt MN, and forming an angle of 90° with the slider.

The individual components appear in their most demonstrative view, i.e. 'handle', 'fork' and 'pitarion' in front view and 'bolt' and 'snake' in side view. In spite of the schematisation, details as the forked end of the 'bolt', the tenons in the 'pitarion' and the curved section of the 'little snake' are readily apparent.

III

Let what are called field-frames²³ be constructed in the following way.

For having made 4 iron bars²⁴, each having a length of 10 ½ digits, a breadth of a little more than two thirds of digit, a thickness so that they cannot be easily bent.²⁵

Let them be named AB , $\Gamma\Delta$, EZ , $H\Theta$, shaped as illustrated below, having rings attached to them $K\Lambda$, MN , ΞO , ΠP , which are 2 digits broad, 1 digit wide²⁶, the same thickness as the bars.

Let the distance between the bars be 3 ½ digits.

Let elements in the form of a ‘pi’ also be made, Σ , T , Y , Φ , X , Ψ , Ω , $\text{‘}A$, attached to the bars AB , $\Gamma\Delta$, EZ , MN , having the same width and thickness as the bars, a breadth²⁷ of two thirds of a digit.

Let also light bronze cylinders²⁸ be, $\text{‘}B$ $\text{‘}T$, $\text{‘}\Delta$ $\text{‘}E$, $\text{‘}\zeta$ $\text{‘}Z$, $\text{‘}H$ $\text{‘}\Theta$, each having a length of two digits, the same thickness as the bars, the diameter of the breadth 1 ⅓ digits.

Let them also have rings attached round their convex surface $M^A M^B$, $M^T M^A$, $M^E M^E$, $M^Z M^H$, being 1 ¼ digits away from $\text{‘}B$, $\text{‘}\Delta$, $\text{‘}\zeta$, $\text{‘}H$; let them have a width of two thirds of a digit, the same thickness as the bars.

Let the cylinders $\text{‘}B$ $\text{‘}T$, $\text{‘}\Delta$ $\text{‘}E$, $\text{‘}\zeta$ $\text{‘}Z$, $\text{‘}H$ $\text{‘}\Theta$, have along the diameter incisions $\zeta, \zeta, < \zeta, \zeta >$, into which are tightly slotted on their narrow edges bars²⁹ M^O , M^O , M^O , M^O , each having a length of 3 digits, a width of two thirds of a digit.

²³ *Kambestria*, sing. *kambestrion*. They correspond to the half-springs in a stone-shooting palintone.

²⁴ Notice that the author uses the same word ‘*kanon*’ to design the wooden boards in the first section and the iron bars in the following ones.

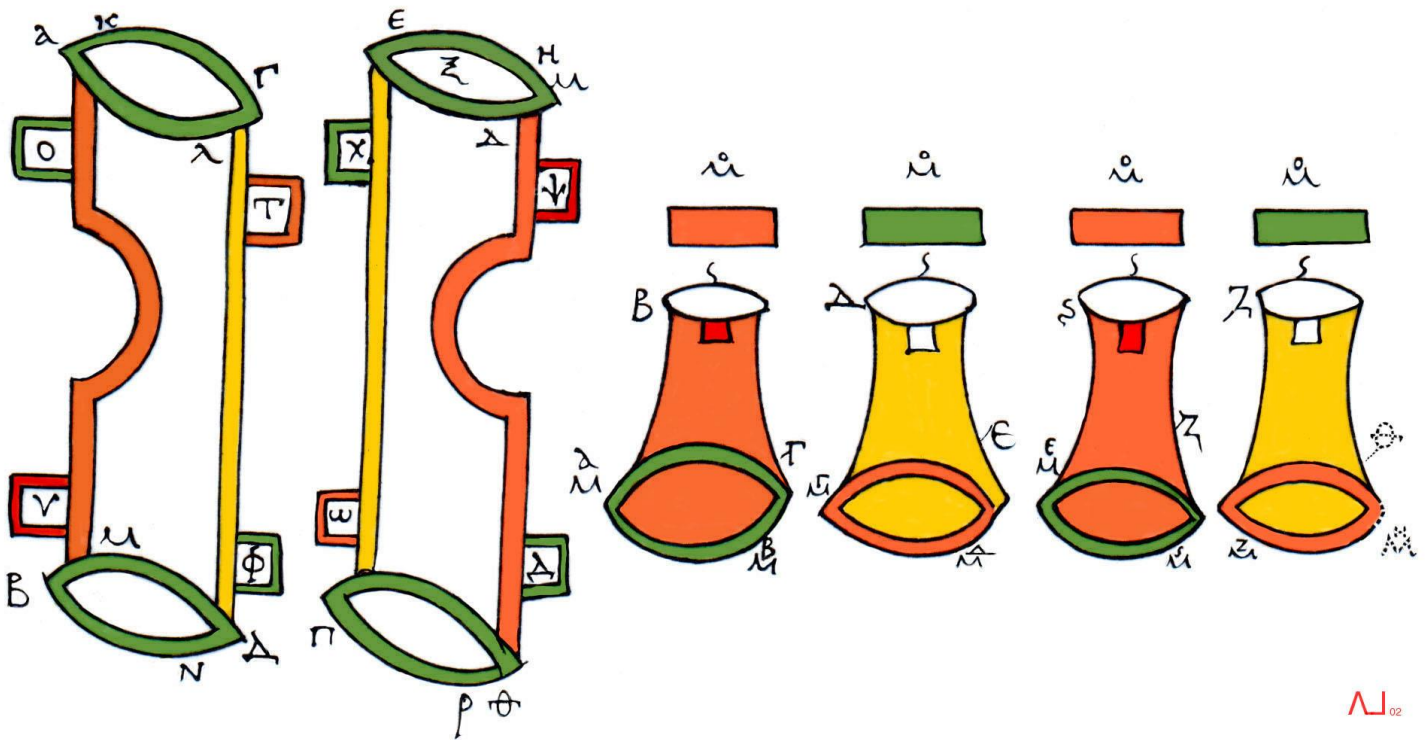
²⁵ A lacuna has been supposed here by Schneider.

²⁶ The ‘breadth’ *ευρος* designs the inner opening of the ring and *πλάτος*, the ‘width’ of the solid part of the ring.

²⁷ See the preceding footnote.

²⁸ The washers.

²⁹ The levers.



The third diagram is devoted to the two 'field-frames', to the left, and their corresponding four 'cylinders' (washers) and four 'bars' (levers), arranged as a row to the right.

The representation is a kind of semi-perspective, with the circles rendered as pointed ellipses. The washers show clearly the slots for the levers. Instead of being cylinders with attached rims, as described in the text, they rather look like concave-side cone frustums.

In each 'field-frame', one of the bars possesses a noticeable bulge in the middle of its length, a feature unmentioned in the text. It has been rotated 90° to make it more apparent. A 'pitarion' is inserted near the upper and lower ends of each bar. All along the diagrams, the pointed ellipses are always placed with their long axe perpendicularly set to the longitudinal axe of the figure, but in the 'field-frames', the ellipses depicting the rings are set obliquely to the bars, I think that it has been made to mean that the bars are placed offset in relation to the rings, not radially.

IV

Let also what is called the little arch³⁰ be made, shaped as illustrated below, $AB\Gamma\Delta EZH$, having ΓE one foot and 7 ½ digits, the interval formed by the arch ΘK , 5 digits. Let the length of each of the portions A, Z be 4 digits, of each of B, H , 2 digits. Let the gap between A, B and Z, H be about 3 ½ digits. Let it have the same thickness as the above-mentioned bars³¹.

Let what is called the little ladder³², $AMNEO\Pi P\Sigma$, be. (Consisting) of two bars shaped as illustrated below. Let the bar $O\Pi P\Sigma$ have a length of 1 foot and 10 digits, $AMNE$, of 1 foot and 8 digits. Let the width at the portions Y, T be 2 digits; at $\angle AM, NZ$, $O\Pi, P\Sigma$, 1 ¼ digits. Let the thickness...³³ of each of the tenons $A'B, N'T, O'\Delta, P'E$ be 2 digits.

Let the bars $AMNE, O\Pi P\Sigma$ be divided at three equal (points) $\Phi, T, \Psi, X, Y, \Omega$. Let holes be bored, quadrangular holes longitudinally at T, Y ; round holes at Φ, X, Ψ, Ω . And let a cross-piece TY be made, having a length, without the tenons, of 3 digits, a width of 2 ½ digits. Let also be pilasters $\Phi X, \Psi\Omega$, having a length, without the tenons, of 3 digits, a width of 2 ½ digits.

And let the pilasters and cross-piece be inserted into the holes in the bars, and let the tenons of the cross-piece be nailed on their ends to the bars, so that the bars are kept linked and the distance between them is 3 digits.

In addition, let the pieces³⁴ $\varsigma, \varsigma, \angle\varsigma, \varsigma>$ be nailed to the bar AN and to the OP , on both (sides) of the cross-piece TY ; let them have a length of 3 digits³⁵, a width of 1 digit, a suitable thickness. Let them be bored down the middle and be 2 ½ digits apart from one another.

³⁰ *Kamarion*. The frame's upper strut.

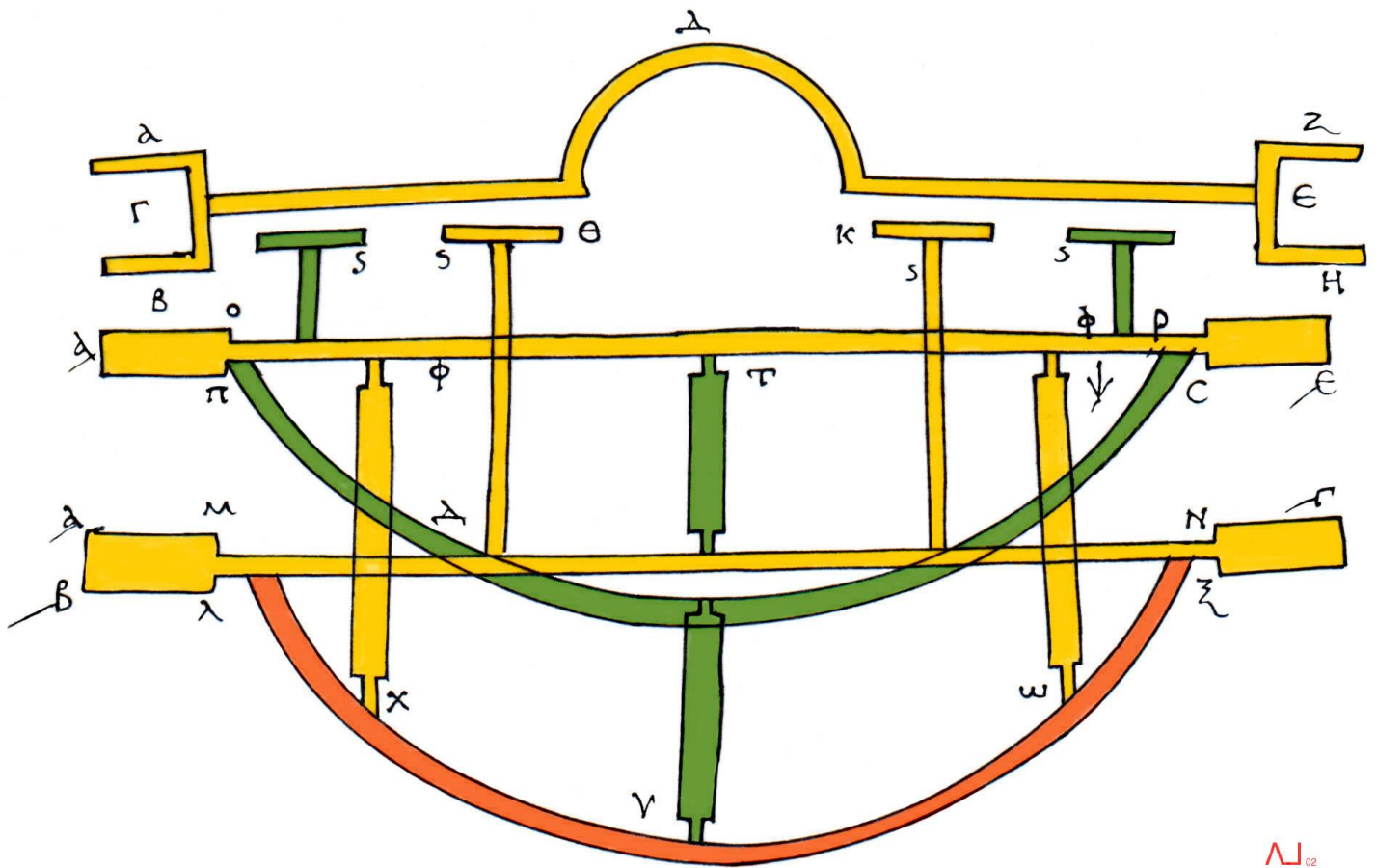
³¹ Those of the field-frames.

³² *Klimakion*. The frame's lower strut.

³³ Here a clear lacuna was detected by Vincent. Surely the copyist skipped one line. Perhaps it could be reconstructed as follows: 'Let their thickness be as the above-mentioned bars. Let the length of each of the tenons $A'B, N'T, O'\Delta, P'E$ be 2 digits'.

³⁴ The clamps that link the case to the ladder.

³⁵ As appears in the manuscript F, one digit in M. One digit also in P and V, but the copyist has skipped the words referring to the width of the clamps, so they cannot be used to support either reading.



The fourth diagram depicts the 'little arch' and the 'little ladder'. The 'little arch' is very clearly drawn, their forked ends are turned 90° to show them frontally.

The 'little ladder' is surely the *Cheiroballistra's* least understandable illustration. Their two main components, instead of look like simple rods tapering towards their ends, have been split in two, one curved bar joining another straight one. To make things worse, the central crosspiece TY has been duplicated. The four T-clamps σ, σ, σ, σ, joining the case to the ladder, are represented of different length, but only to underline that the longer pair stems from the lower bar.

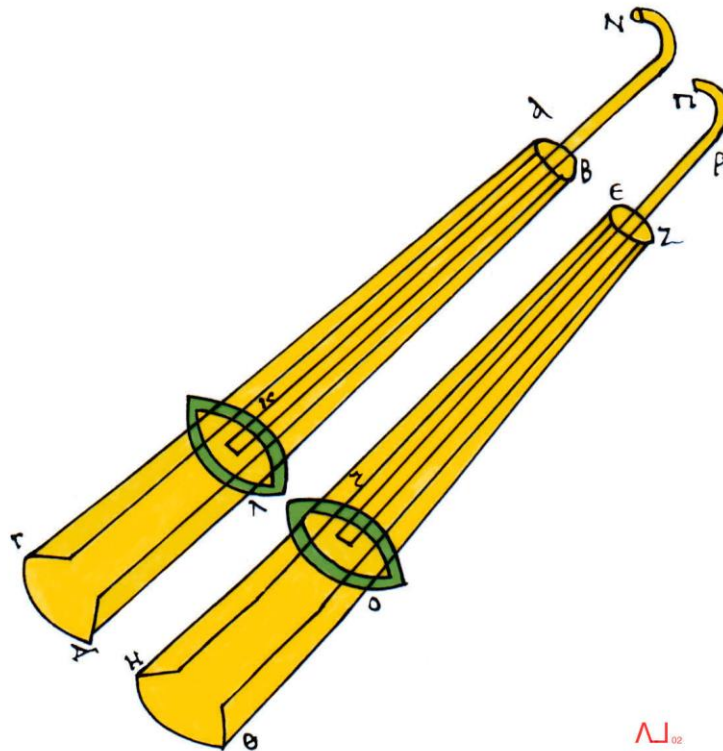
Neither the forked tenons in the 'little arch', nor the bars in the 'little ladder' show any traces of the length differences stated in the text.

V

Let also two conical (elements) $AB\Gamma\Delta$, $EZH\Theta$ be made, each having a length of 11 digits. Let the thickness of the ends AB , EZ of each conical (element) be half a digit, the thickness of each of the bases $\Gamma\Delta$, $H\Theta$, one digit.

Let them have longitudinally quadrilateral grooves, and tenons in the ends AB , EZ , so that when bars are made attached to rings, fitting to the tenons and the grooves, they can be pulled out along the tenons and grooves in the conical (elements) that have been made.

Let the bars attached to rings be $KAMN$, ϵOHP , the rings, KA , ϵO . Let the bars have at their ends hooks MN and ιP , having a height of half a digit.³⁶



The fifth and last diagram pictures the two 'conical objects' (arms), obliquely set. The bars are inserted into the cones' grooves. Only the borders of the grooves are figured, not the bottoms. The rings attached to the bars appear as thin pointed ellipses, the hooks are depicted as simple bends at the ends of the bars.

³⁶ From Wescher -who followed Vincent's guess- on, it has been universally accepted that the text is incomplete and breaks off here abruptly without giving the length, width and thickness of the bars. I shall not deny it, but I must remark that, in the logical descriptive sequence followed in all the *Cheiroballistra* sections, those supposedly missing dimensions of the bars should have been given before the height of their hooks, which is not the case. Therefore, perhaps the section itself is complete.

THE DIAGRAMS AND THE TRANSMISSION OF THE MANUSCRIPTS

The *Cheiroballistra*, as most -if not all- technical treatises, was furnished from the beginning with a set of illustrations. Such illustrations were not a mere embellishment, like it was the case in poetry, novel or history books, but they were real practical diagrams, with little or no concessions to aesthetics.

In the *Cheiroballistra*'s original edition -perhaps sometime around the beginning of the second century A.D.- diagrams were interspersed amongst the text sections -written in Greek- along a papyrus scroll³⁷. The figures belonged to a kind of technical drawing totally alien to us now, but then perfectly familiar to Greek and Roman architects and engineers. They could include, all in one diagram, the plan (*ichnographia*) and side-elevations (*ortographias*) of the object. Some kind of frontal rudimentary perspective (*scenographia*) was also employed³⁸. Furthermore, the Classical draughtsman considered that it was legitimate to turn parts through 90° to make them visible in an illustration. The *Cheiroballistra* diagrams were also lettered according with the text and each component wore its name written below.

We do not know how many times our treatise had been copied from copies made in their turn from the original -and, therefore, text and diagrams subjected to deformation- before a fragmentary, perhaps untitled, copy -written in Greek uncial majuscule letters- reached a Byzantine scriptorium around the tenth century A.D. They had decided to make a compilation of the available Greek treatises on poliorcetic and the fragmentary *Cheiroballistra*³⁹ was included in it. A

³⁷ Some tenuous hints even point at the possibility that the original treatise was written in Latin and afterwards translated into Greek.

³⁸ Vitruvius I, 2, 2 : '*Ichonography* (plan) demands the competent use of compass and rule; by these plans are laid out upon the sites provided. *Orthography* (elevation), however, is the vertical image of the front, and a figure slightly tinted to show the lines of the future work. *Scenography* (perspective) also is the shading of the front and the retreating sides, and the correspondence of all lines to the vanishing point, which is the centre of a circle'.

³⁹ As I have already said, maybe the title '*Heron's Construction and Dimensions of the Cheiroballistra*' was then added to the titleless treatise.

reader dictated the text to several scribes at a time, in order to increase production of copies⁴⁰. One of those copies -written already in Greek cursive- has survived until our times, in the manuscript known in short as **M**.

I have chosen **M**'s diagrams to accompany the translation of the text because, being the oldest surviving copies, they draw us as near as it is possible now to the appearance of the original illustrations. They do not seem, actually, to be very far removed from their archetypes and display strongly archaising characteristics which show us that the conservatism of the Macedonian Renaissance reached even the technical illustrations.

The lettering is still almost complete, but only the captions belonging to the first diagram survive, albeit in an utterly corrupt form⁴¹.

One of the copies made in parallel to **M** and now lost -we shall call it **y**- was transferred to another Byzantine scriptorium, where a new and more ambitious compilation was being prepared, maybe during the eleventh century A.D. We can have a faint idea about how diagrams in **y** looked like through a careless partial copy of it which was made in the sixteenth century A.D., **F**. **F**'s figures, roughly traced and without lettering as they are, resemble very closely those of **M**.

From the several copies made, still in the eleventh century, of the aforesaid compilation -we shall call it **z**- two almost identical ones still survive in good condition, **P** and **V**⁴².

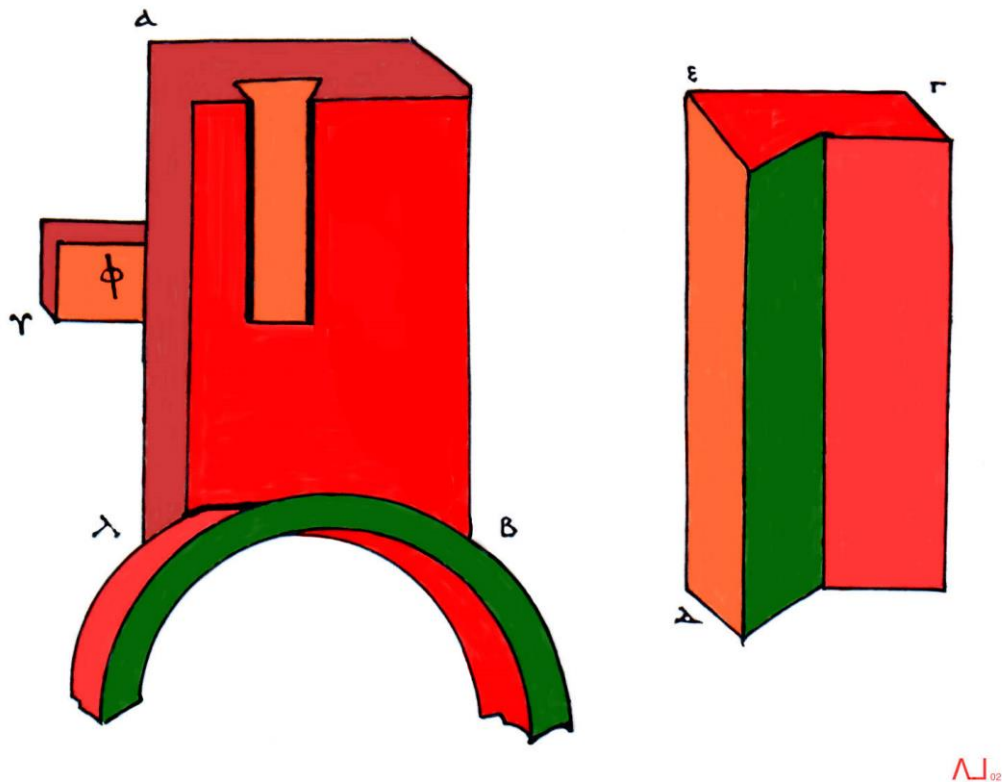
When the new Byzantine compilation was made, the person in charge decided to change the first and last diagrams for two brand new ones. Maybe those illustrations were very deteriorated in **y** and needed thus to be replaced, but I think, especially when we look at the first diagram, that they were simply 'updated' to make them more understandable to contemporary eyes. It is important to remark that whoever decided to make it still understood what he was doing and

⁴⁰ Of course, spaces were left in the text, where later the diagrams were traced by specialist copyists.

⁴¹ **θηλυς αγκων** could be translated as 'female arm (curve)', and **αγκων κατω**, as 'arm (curve) downwards', which is nonsense. Even if it is a palaeographically difficult correction, I am inclined to accept Schneider's (1906, 149, n. 2) proposal of amending the two corrupt labels to **θηλυς κανων** -'female board'- and **αρρην κανων** -'male board'-.

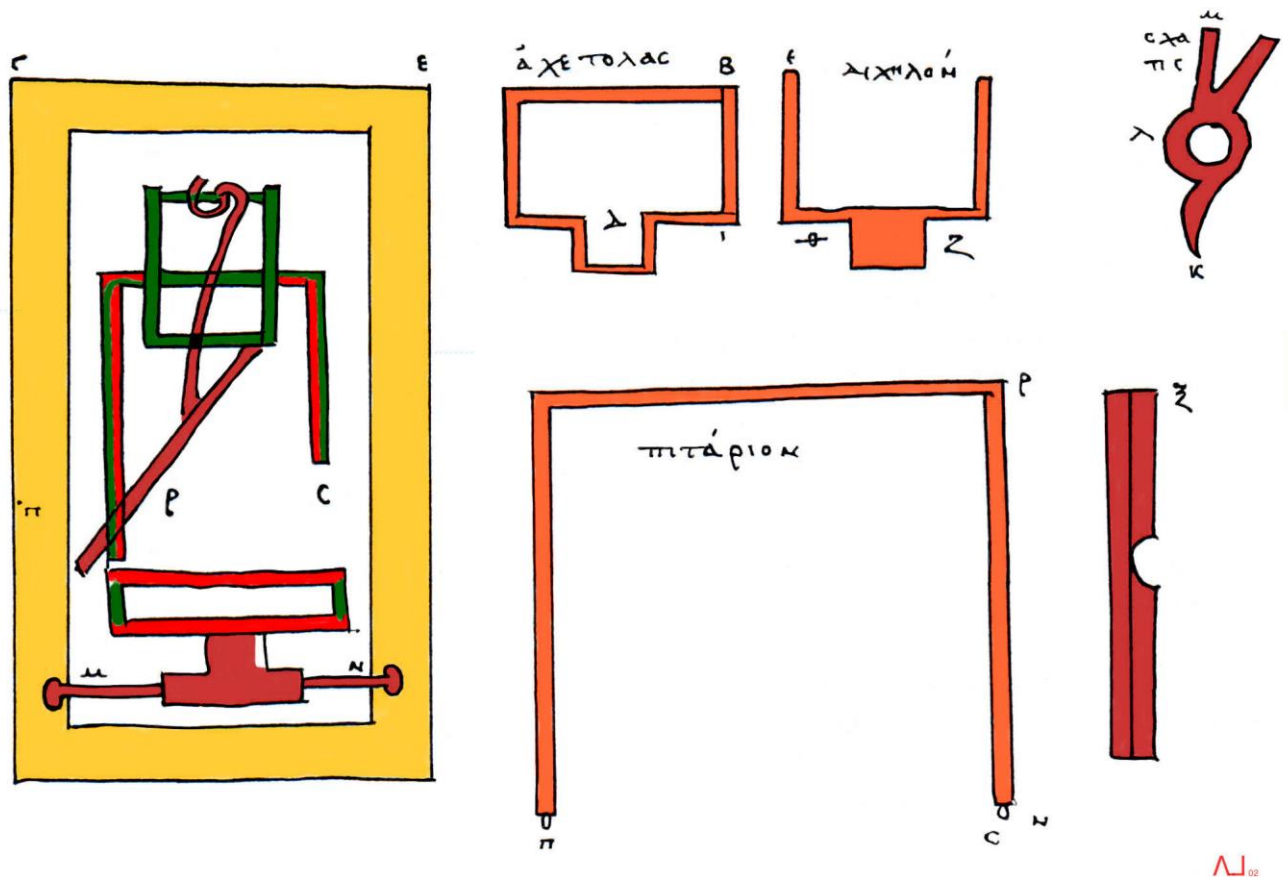
⁴² The two following diagrams have been copied from **P**.

how torsion artillery components looked like. We should not be too surprised about that because there are first-hand literary references until the eleventh century A.D. which prove that some kinds of catapults were still in use by that time.



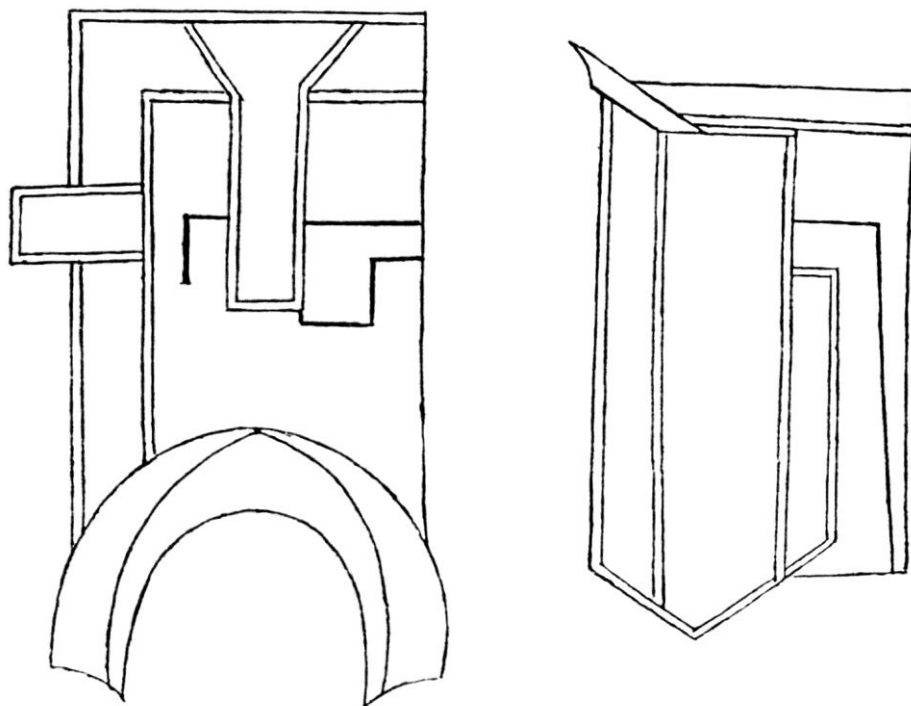
Both, case and slider, have been rendered in some kind of axonometric perspective. The case is, as usually, seen from above, but the projecting block has been turned 90° towards the left to make it more visible. The slider is, apparently, seen from its side. The captions and a good deal of the lettering have disappeared, meanwhile.

Curiously, the second diagram has preserved in **P** and **V** -via **y**, no doubt- the captions, corrupt but still fully identifiable, below each one of the trigger-mechanism's elements.



To finish with the diagrams, I shall try to give an idea -no matter if inaccurate- of how far could be even **M**'s figures from those in the original archetype. Of course, we cannot trace their evolution backwards, but we can check it forwards. I have chosen the first diagram -depicting case and slider- to show the changes. We have already seen that, perhaps no more than a century after **M** was copied, somebody decided to substitute the figure for a completely new one. Let us see now how this new diagram looks like in one of the secondary manuscripts, **E**⁴³, copied in the sixteenth century A.D.

⁴³ It is kept in the library of the Escorial monastery (Madrid, Spain). Escorialensis Φ - II - 22.



The former axonometric figures are now little more than two groups of calligraphical but almost unintelligible lines and no trace of the lettering survives. **E** is only five centuries younger than **V** -from which it descends- and six centuries younger than **M**, while **M** is maybe as much as nine centuries apart from the original. Obviously, the diagrams have deteriorated more steadily in the later period, at the hands of copyists who didn't understand at all what they were drawing, while during the earlier period only carelessness and a deteriorated original could generate significant changes. Anyway, the sequence can help to get an insight into the dynamics of change during the transmission of ancient diagrams.

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